



PAPER ID-310505

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Subject Code: RCA1503

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MCA-INT
(SEM V) THEORY EXAMINATION 2024-25
JAVA PROGRAMMING

TIME: 3 HRS**M.MARKS: 70**

Note: Attempt all Sections. In case of any missing data; choose suitably.

SECTION A**1. Attempt all questions in brief.****2 x 07 = 14**

Q no.	Question	CO	Level
a.	List out the protocols name used for sending and receiving E-Mail?	1	K1
b.	Give name of two protocols commonly used in video conferencing to facilitate real-time communication.	1	K2
c.	Explain types data types available in Java.	2	K2
d.	Describe JDBC-ODBC bridge.	3	K1
e.	Explain the use of JAR files in Java.	4	K2
f.	Explain JSP advantages on Servlet.	5	K2
g.	Summarize data hiding in the reference of Encapsulation.	2	K2

SECTION B**2. Attempt any three of the following:****07 x 3 = 21**

a.	Illustrate E-mail security and why it is necessary?	1	K3
b.	Explain inheritance and types of inheritance in Java.	2	K2
c.	Use Java to implement an event-driven program in Swing to handle button clicks and update a label with the number of clicks in real time.	3	K3
d.	Illustrate different types of EJBs (stateless, stateful, and singleton). How would you choose the appropriate type of EJB for a banking application where multiple users access shared data, and why?	4	K3
e.	Explain how the lifecycle of a Servlet differs from that of a JSP, and describe the role of the init(), service(), and destroy() methods in a Servlet.	5	K2

SECTION C**3. Attempt any one part of the following:****07 x 1 = 07**

a.	Explain various hardware and software requirements of video conferencing?	1	K2
b.	Summarize the following terms in details: 1. Internet Service Provider 2. Backbone Service Provider 3. VOIP	1	K2

4. Attempt any one part of the following:**07 x 1 = 07**

a.	Differentiate abstract class with interface with proper example.	2	K2
b.	Explain threads and its life cycle with proper diagram.	2	K2

5. Attempt any one part of the following:**07 x 1 = 07**

a.	Use JDBC program to display record from MySQL database table named "Employee" where student_id is greater than 100 and less than 231.	3	K3
b.	Demonstrate event delegation process in Java and give an example of any event like mouse event, keyboard event and windows events.	3	K3



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JAVA PROGRAMMING

TIME: 3 HRS**M.MARKS: 70****6. Attempt any one part of the following:****07 x 1 = 07**

a.	Describe the role of the container in EJB. How does it manage the lifecycle of Enterprise Beans?	4	K2
b.	Explain Remote Method Invocation (RMI), and how does it facilitate communication between distributed Java applications	4	K2

7. Attempt any one part of the following:**07 x 1 = 07**

a.	Implement a Servlet that tracks a user's login session when the user logs in, store the user information (such as username), and display a welcome message when the user revisits the page.	5	K3
b.	Use Cookie to write a servlet program which sets some cookie on the client machine that must be expires within 2 hours.	5	K3

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